

Electrical Transient Analyzer Program

Short-Circuit Analysis

IEC 60909 Standard

3-Phase, LG, LL, & LLG Fault Currents

	Swing	V-Control	Load	Total			
Number of Buses:	1	0	14	15			
	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	12	0	1	14
	Synchronous Generator	Power Grid	Synchronous Motor	Induction Machines	Lumped Load	Total	
Number of Machines:	1	0	0	4	0	5	

Project:

Location:

Contract:

Engineer:

Filename:

ETAP

20.6.0C

Study Case: SC

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Date: 30-03-2026

SN:

Revision: Base

Config.: Normal

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus AUX	Load	0.400	0.410	1	100.00	-30.00
Bus Crane	Load	0.400	0.410	1	100.00	-30.00
Bus Gel Breaking	Load	0.400	0.410	1	100.00	-30.00
Bus GTG	SWNG	6.000	6.000	1	100.00	0.00
Bus Heater	Load	0.400	0.410	1	100.00	-30.00
Bus HT	Load	0.400	0.410	1	100.00	-30.00
Bus Lighting P.	Load	0.400	0.410	1	100.00	-30.00
Bus Mkomp	Load	6.000	6.000	1	100.00	0.00
Bus P. Trafo	Load	6.000	6.000	1	100.00	0.00
Bus S. Trafo	Load	0.400	0.410	1	100.00	-30.00
Bus Sea Water	Load	0.400	0.410	1	100.00	-30.00
Bus UPS	Load	0.400	0.410	1	100.00	-30.00
Gen Bus Power Supply	Load	6.000	6.000	1	100.00	0.00
LV SW2	Load	0.400	0.410	1	100.00	-30.00
LV Switchrack	Load	0.400	0.410	1	100.00	-30.00

15 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP.

Line/Cable/Busway Input Data

ohms or siemens per 1000 m per Conductor (Cable) or per Phase (Line/Busway)												
Line/Cable/Busway												
ID	Library	Size	Length		#/Phase	T (°C)	R1	X1	Y1	R0	X0	Y0
			Adj. (m)	% Tol.								
Cable AUX SP	0,6NCUS3	1.5	100.0	0.0	1	75	14.6881400	0.1000000	0.0000208	23.3541400	0.2540000	
Cable Kompessor	10NCUS3	35	150.0	0.0	1	75	0.6371217	0.1120000		1.0130050	0.2844800	
Cable Motor Gel breaking	1,0NCUS3	70	30.0	0.0	1	75	0.3261911	0.0726000		0.5186438	0.1844000	
Cable1	1,0NCUS3	35	70.0	0.0	1	75	0.6380755	0.0743000		1.0145400	0.1887200	
Cable2	1,0NCUS3	25	70.0	0.0	1	75	0.8841494	0.0768000		1.4057980	0.1950700	
Cable3	1,0NCUS3	120	90.0	0.0	1	75	0.1869399	0.0705000		0.2972344	0.1790700	
Cable4	1,0NCUS3	25	50.0	0.0	1	75	0.8841494	0.0768000		1.4057980	0.1950700	
Cable5	1,0NCUS3	25	30.0	0.0	1	75	0.8841494	0.0768000		1.4057980	0.1950700	
Cable6	1,0NCUS3	25	30.0	0.0	1	75	0.8841494	0.0768000		1.4057980	0.1950700	
Cable8	1,0NCUS3	400	30.0	0.0	1	75	0.0610416	0.0694000		0.0970562	0.1762800	
Cable9	10NCUS3	25	30.0	0.0	1	75	0.8841494	0.1170000		1.4057690	0.2971800	
Cable11	10NCUS3	95	500.0	0.0	1	75	0.2355824	0.0955000		0.3745761	0.2425700	

Line / Cable / Busway resistances are listed at the specified temperatures.

2-Winding Transformer Input Data

Transformer	Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	MVA	Prim. kV	Sec. kV	% Z	X/R	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
Power Trafo	0.500	6.000	0.400	4.00	1.50	0	0	0	-2.500	0	4.00	Dyn	30.00

2-Winding Transformer Grounding Input Data

Transformer	Rating			Grounding									
	ID	MVA	Prim. kV	Sec. kV	Conn.	Primary				Secondary			
					Type	Type	kV	Amp	ohm	Type	kV	Amp	ohm
Power Trafo		0.500	6.000	0.400	D/Y					Solid			

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
Power Trafo	2W XFMR	Bus P. Trafo	Bus S. Trafo	412.56	618.83	743.75	
Cable AUX SP	Cable	LV SW2	Bus AUX	87268.19	594.14	87270.21	0.0000003
Cable Kompessor	Cable	Gen Bus Power Supply	Bus Mkomp	26.55	4.67	26.95	
Cable Motor Gel breaking	Cable	LV SW2	Bus Gel Breaking	581.41	129.40	595.64	
Cable1	Cable	LV Switchrack	Bus Heater	2653.75	309.01	2671.68	
Cable2	Cable	LV Switchrack	Bus Lighting P.	3677.16	319.41	3691.01	
Cable3	Cable	LV Switchrack	Bus Crane	999.62	376.98	1068.34	
Cable4	Cable	LV Switchrack	Bus Sea Water	2626.55	228.15	2636.44	
Cable5	Cable	LV Switchrack	Bus HT	1575.93	136.89	1581.86	
Cable6	Cable	LV Switchrack	Bus UPS	1575.93	136.89	1581.86	
Cable8	Cable	Bus S. Trafo	LV Switchrack	108.80	123.70	164.74	
Cable9	Cable	Gen Bus Power Supply	Bus P. Trafo	7.37	0.98	7.43	
Cable11	Cable	Bus GTG	Gen Bus Power Supply	32.72	13.26	35.31	
CB21	Tie Breakr	LV Switchrack	LV SW2				

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Synchronous Generator Input Data

Synchronous Generator						% Impedance in Machine Base								
						Rating				Xd''			Grounding	
ID	Type	MVA	kV	RPM	% PF	R	Adj.	Tol.	R/X	Xd, sat	Conn.	Type	Amp	Type
GTG	Steam Turbo	1.111	6.000	1500	90.00	1.000	19.00	0.0	0.05	155.00	Wye	Solid	0.00	Turbine 130%
Total Connected Synchronous Generators (= 1,00) : 1,111 MVA														

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Induction Machine Input Data

Induction Machine			Connected Bus	Rating					Pos. Seq. Imp. (Motor Base)			Grounding			
ID	Type	Qty	ID	HP/kW	kVA	kV	Amp	PF	% R	% X"	R/X"	Conn.	Type	Amp	MW/PP
Motor Kompessor	Motor	1	Bus Mkomp	300.00	355.89	6.000	34.25	89.79	3.52	13.64	0.26	Wye	Open		0.15
Pedestal Crane	Motor	1	Bus Crane	90.00	119.89	0.400	173.05	91.50	4.67	16.00	0.29	Wye	Open		0.05
Gel Breaking (F)	Motor	1	Bus Gel Breaking	110.00	141.51	0.400	204.25	91.58	4.41	16.07	0.27	Wye	Open		0.06
Sea Water	Motor	1	Bus Sea Water	11.00	16.45	0.400	23.75	87.47	8.32	14.44	0.58	Wye	Open		0.01

Total Connected Induction Machines (= 4): 633.7 kVA

SHORT-CIRCUIT REPORT

Fault at bus: **Bus AUX**
Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus AUX	Total	0.00	0.164	0.00	114.35	114.78	0.137	0.137	8.78E+004	2.04E+003	1.39E+005	2.44E+003
LV SW2	Bus AUX	104.37	0.164	104.44	104.83	105.20	0.137	0.137	8.78E+004	2.04E+003	1.39E+005	2.44E+003
Bus Gel Breaking	LV SW2	104.46	0.022	104.49	104.85	105.21	0.011	0.000	3.55E+003	1.09E+004		
Bus Heater	LV Switchrack	104.37	0.000	104.44	104.83	105.20	0.000	0.000				
Bus Lighting P.	LV Switchrack	104.37	0.000	104.44	104.83	105.20	0.000	0.000				
Bus Crane	LV Switchrack	104.50	0.018	104.51	104.86	105.21	0.010	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	104.41	0.003	104.46	104.84	105.20	0.001	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	104.37	0.000	104.44	104.83	105.20	0.000	0.000				
Bus UPS	LV Switchrack	104.37	0.000	104.44	104.83	105.20	0.000	0.000				
Bus S. Trafo	LV Switchrack	104.46	0.122	104.55	104.75	105.24	0.115	0.137	6.85E+002	1.96E+003	5.86E+002	9.33E+002

		3-Phase	L-G	L-L	L-L-G
Initial Symmetrical Current (kA, rms)	:	0.164	0.137	0.142	0.154
Peak Current (kA), Method C	:	0.237	0.198	0.205	0.222
Breaking Current (kA, rms, symm)	:		0.137	0.142	0.154
Steady State Current (kA, rms)	:	0.164	0.137	0.142	0.154

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: **Bus Crane**
Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus Crane	Total	0.00	6.682	0.00	102.94	119.77	5.942	5.942	1.22E+003	1.77E+003	2.17E+003	1.89E+003
LV Switchrack	Bus Crane	44.05	5.658	53.15	97.79	110.62	5.378	5.942	1.57E+003	2.00E+003	2.17E+003	1.89E+003
Pedestal Crane	Bus Crane	105.00	1.090	105.00	105.00	105.00	0.602	0.000	3.70E+003	1.27E+004		
Bus Heater	LV Switchrack	44.05	0.000	53.15	97.79	110.62	0.000	0.000				
Bus Lighting P.	LV Switchrack	44.05	0.000	53.15	97.79	110.62	0.000	0.000				
Bus Sea Water	LV Switchrack	45.96	0.100	54.05	97.94	111.19	0.048	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	44.05	0.000	53.15	97.79	110.62	0.000	0.000				
Bus UPS	LV Switchrack	44.05	0.000	53.15	97.79	110.62	0.000	0.000				
Bus S. Trafo	LV Switchrack	49.12	4.711	60.76	95.31	109.94	4.855	5.942	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	44.05	0.000	53.15	97.79	110.62	0.000	0.000				
Bus Gel Breaking	LV SW2	47.69	0.850	55.16	97.77	111.63	0.476	0.000	3.55E+003	1.09E+004		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)	:	6.682		5.942		5.607		6.772				
Peak Current (kA), Method C	:	11.118		9.887		9.330		11.267				
Breaking Current (kA, rms, symm)	:	5.942		5.942		5.607		6.772				
Steady State Current (kA, rms)	:	2.534		5.942		5.607		6.772				

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: Bus Gel Breaking

Nominal kV = 0.400

Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus Gel Breaking	Total	0.00	7.943	0.00	96.08	118.59	7.663	7.663	9.16E+002	1.57E+003	1.51E+003	1.26E+003
LV SW2	Bus Gel Breaking	29.12	6.709	37.21	95.61	111.28	6.923	7.663	1.17E+003	1.80E+003	1.51E+003	1.26E+003
Gel Breaking (F)	Bus Gel Breaking	105.00	1.287	105.00	105.00	105.00	0.774	0.000	2.97E+003	1.08E+004		
Bus AUX	LV SW2	29.12	0.000	37.21	95.61	111.28	0.000	0.000				
Bus Heater	LV Switchrack	29.12	0.000	37.21	95.61	111.28	0.000	0.000				
Bus Lighting P.	LV Switchrack	29.12	0.000	37.21	95.61	111.28	0.000	0.000				
Bus Crane	LV Switchrack	35.73	0.856	41.24	95.74	113.15	0.519	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	31.45	0.121	38.42	95.76	112.03	0.063	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	29.12	0.000	37.21	95.61	111.28	0.000	0.000				
Bus UPS	LV Switchrack	29.12	0.000	37.21	95.61	111.28	0.000	0.000				
Bus S. Trafo	LV Switchrack	34.94	5.734	46.36	92.67	110.35	6.341	7.663	6.85E+002	1.96E+003	5.86E+002	9.33E+002
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)	:	7.943		7.663		6.650		8.535				
Peak Current (kA), Method C	:	13.683		13.200		11.455		14.702				
Breaking Current (kA, rms, symm)	:			7.663		6.650		8.535				
Steady State Current (kA, rms)	:	2.606		7.663		6.650		8.535				

Indicates a fault current contribution from a three-winding transformer.

* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: **Bus GTG**
Nominal kV = 6.000
Voltage c Factor = 1.10 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus GTG	Total	0.00	1.018	0.00	104.94	104.12	1.140	1.140	1.64E+002	1.03E+003	9.14E+001	6.40E+002
Gen Bus Power Supply GTG	Bus GTG	1.53	0.418	1.04	104.76	104.68	0.285	0.000	7.81E+002	2.41E+003		
	Bus GTG	110.00	0.608	110.00	110.00	110.00	0.859	1.140	9.14E+001	1.74E+003	9.14E+001	6.40E+002
Bus Mkomp	Gen Bus Power Supply	2.26	0.264	1.53	104.63	104.94	0.177	0.000	1.02E+003	3.84E+003		
Bus P. Trafo	Gen Bus Power Supply	1.65	0.154	1.13	104.74	104.72	0.108	0.000	2.51E+003	6.32E+003		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:		1.018		1.140		0.857		1.114		
Peak Current (kA), Method C		:		2.413		2.703		2.032		2.640		
Breaking Current (kA, rms, symm)		:		1.140		0.857		1.114				
Steady State Current (kA, rms)		:		0.195		1.140		0.857		1.114		

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: **Bus Heater**
Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus Heater	Total	0.00	3.982	0.00	106.12	117.91	3.496	3.496	3.16E+003	1.76E+003	4.81E+003	1.72E+003
LV Switchrack	Bus Heater	77.54	3.982	81.78	100.27	109.03	3.496	3.496	3.16E+003	1.76E+003	4.81E+003	1.72E+003
Bus Lighting P.	LV Switchrack	77.54	0.000	81.78	100.27	109.03	0.000	0.000				
Bus Crane	LV Switchrack	80.87	0.440	83.63	100.74	109.67	0.242	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	78.72	0.062	82.35	100.45	109.34	0.029	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	77.54	0.000	81.78	100.27	109.03	0.000	0.000				
Bus UPS	LV Switchrack	77.54	0.000	81.78	100.27	109.03	0.000	0.000				
Bus S. Trafo	LV Switchrack	80.21	2.949	85.32	98.34	109.48	2.932	3.496	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	77.54	0.000	81.78	100.27	109.03	0.000	0.000				
Bus Gel Breaking	LV SW2	79.85	0.532	83.05	100.49	109.54	0.293	0.000	3.55E+003	1.09E+004		

		3-Phase	L-G	L-L	L-L-G
Initial Symmetrical Current (kA, rms)	:	3.982	3.496	3.415	3.962
Peak Current (kA), Method C	:	5.772	5.067	4.949	5.742
Breaking Current (kA, rms, symm)	:		3.496	3.415	3.962
Steady State Current (kA, rms)	:	2.371	3.496	3.415	3.962

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer.

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Fault at bus:	Bus HT
Nominal kV	= 0.400
Voltage c Factor	= 1.05 (User-Defined)

Positive & Zero Sequence Impedances												
Contribution		3-Phase Fault		Line-To-Ground Fault					Looking into "From Bus"			
From Bus	To Bus	% V	kA	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
ID	ID	From Bus	Symm. rms	Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus HT	Total	0.00	5.502	0.00	100.97	119.01	5.031	5.031	2.08E+003	1.58E+003	3.09E+003	1.28E+003
LV Switchrack	Bus HT	63.43	5.502	69.56	98.23	110.29	5.031	5.031	2.08E+003	1.58E+003	3.09E+003	1.28E+003
Bus Heater	LV Switchrack	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus Lighting P.	LV Switchrack	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus Crane	LV Switchrack	67.99	0.608	72.21	98.76	111.31	0.349	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	65.05	0.086	70.37	98.45	110.76	0.042	0.000	5.07E+004	8.37E+004		
Bus UPS	LV Switchrack	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus S. Trafo	LV Switchrack	67.06	4.075	74.54	95.57	110.67	4.220	5.031	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus Gel Breaking	LV SW2	66.60	0.735	71.39	98.45	111.08	0.421	0.000	3.55E+003	1.09E+004		

		3-Phase	L-G	L-L	L-L-G
Initial Symmetrical Current (kA, rms)	:	5.502	5.031	4.690	5.681
Peak Current (kA), Method C	:	8.094	7.401	6.899	8.357
Breaking Current (kA, rms, symm)	:		5.031	4.690	5.681
Steady State Current (kA, rms)	:	2.511	5.031	4.690	5.681

Indicates a fault current contribution from a three-winding transformer.

* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer.

Fault at bus:

Nominal kV

Voltage c Factor

Bus Lighting P.

= 0.400

= 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus Lighting P.	Total	0.00	3.171	0.00	107.98	117.70	2.740	2.740	4.19E+003	1.77E+003	6.43E+003	1.74E+003
LV Switchrack	Bus Lighting P.	85.30	3.171	88.41	101.32	108.38	2.740	2.740	4.19E+003	1.77E+003	6.43E+003	1.74E+003
Bus Heater	LV Switchrack	85.30	0.000	88.41	101.32	108.38	0.000	0.000				
Bus Crane	LV Switchrack	87.92	0.350	89.85	101.75	108.83	0.190	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	86.23	0.050	88.85	101.47	108.62	0.023	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	85.30	0.000	88.41	101.32	108.38	0.000	0.000				
Bus UPS	LV Switchrack	85.30	0.000	88.41	101.32	108.38	0.000	0.000				
Bus S. Trafo	LV Switchrack	87.36	2.348	91.05	99.77	108.87	2.298	2.740	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	85.30	0.000	88.41	101.32	108.38	0.000	0.000				
Bus Gel Breaking	LV SW2	87.12	0.424	89.40	101.54	108.75	0.229	0.000	3.55E+003	1.09E+004		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:		3.171		2.740		2.728		3.117		
Peak Current (kA), Method C		:		4.578		3.956		3.938		4.500		
Breaking Current (kA, rms, symm)		:		2.740		2.740		2.728		3.117		
Steady State Current (kA, rms)		:		2.253		2.740		2.728		3.117		

Indicates a fault current contribution from a three-winding transformer.

* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer.

Fault at bus: **Bus Mkomp**
Nominal kV = 6.000
Voltage c Factor = 1.10 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus Mkomp	Total	0.00	1.015	0.00	103.24	108.09	1.111	1.111	1.85E+002	1.03E+003	1.86E+002	6.86E+002
Gen Bus Power Supply	Bus Mkomp	2.10	0.749	3.27	103.70	107.18	0.936	1.111	2.13E+002	1.40E+003	1.86E+002	6.86E+002
Motor Kompessor	Bus Mkomp	110.00	0.267	110.00	110.00	110.00	0.175	0.000	9.90E+002	3.83E+003		
Bus P. Trafo	Gen Bus Power Supply	2.21	0.155	3.35	103.68	107.23	0.106	0.000	2.51E+003	6.32E+003		
Bus GTG	Gen Bus Power Supply	4.29	0.600	7.26	104.04	105.76	0.832	1.111	1.24E+002	1.75E+003	1.43E+002	6.74E+002
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:		1.015		1.111		0.853		1.109		
Peak Current (kA), Method C		:		2.337		2.558		1.963		2.555		
Breaking Current (kA, rms, symm)		:		1.111		1.111		0.853		1.109		
Steady State Current (kA, rms)		:		0.195		1.111		0.853		1.109		

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer.

Fault at bus: **Bus P. Trafo**
Nominal kV = 6.000
Voltage c Factor = 1.10 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus P. Trafo	Total	0.00	1.017	0.00	103.85	106.94	1.119	1.119	1.77E+002	1.03E+003	1.55E+002	6.76E+002
Gen Bus Power Supply	Bus P. Trafo	0.67	0.865	0.96	103.98	106.71	1.014	1.119	1.64E+002	1.21E+003	1.55E+002	6.76E+002
Bus S. Trafo	Bus P. Trafo	12.35	0.156	68.16	63.37	115.93	0.107	0.000	2.50E+003	6.32E+003		
Bus Mkomp	Gen Bus Power Supply	1.41	0.266	1.45	103.86	106.97	0.175	0.000	1.02E+003	3.84E+003		
Bus GTG	Gen Bus Power Supply	2.87	0.602	5.00	104.34	105.28	0.839	1.119	1.24E+002	1.75E+003	1.43E+002	6.74E+002
LV Switchrack	Bus S. Trafo	15.07	2.279	69.45	63.68	116.06	1.388	0.000	1.41E+003	6.10E+003		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:	1.017	1.119		0.855		1.108				
Peak Current (kA), Method C		:	2.363	2.600		1.986		2.574				
Breaking Current (kA, rms, symm)		:		1.119		0.855		1.108				
Steady State Current (kA, rms)		:	0.195	1.119		0.855		1.108				

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: **Bus S. Trafo**

Nominal kV = 0.400

Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus S. Trafo	Total	0.00	9.877	0.00	91.50	105.05	11.555	11.555	4.53E+002	1.39E+003	4.13E+002	6.19E+002
LV Switchrack	Bus S. Trafo	2.85	2.374	2.08	91.61	105.81	1.729	0.000	2.09E+003	5.70E+003		
Bus P. Trafo	Bus S. Trafo	39.66	7.504	60.67	104.46	76.29	9.826	11.555 *	5.76E+002	1.83E+003	4.13E+002	6.19E+002
Bus Heater	LV Switchrack	2.85	0.000	2.08	91.61	105.81	0.000	0.000				
Bus Lighting P.	LV Switchrack	2.85	0.000	2.08	91.61	105.81	0.000	0.000				
Bus Crane	LV Switchrack	10.48	1.012	7.69	90.63	108.69	0.743	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	5.39	0.144	3.63	91.49	106.90	0.090	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	2.85	0.000	2.08	91.61	105.81	0.000	0.000				
Bus UPS	LV Switchrack	2.85	0.000	2.08	91.61	105.81	0.000	0.000				
Bus AUX	LV SW2	2.85	0.000	2.08	91.61	105.81	0.000	0.000				
Bus Gel Breaking	LV SW2	7.76	1.223	5.69	90.65	107.82	0.897	0.000	3.55E+003	1.09E+004		
Gen Bus Power Supply	Bus P. Trafo	39.92	0.513	60.59	104.45	76.56	0.356	0.000	1.00E+003	6.79E+002		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:	9.877	11.555		8.264		12.036				
Peak Current (kA), Method C		:	19.543	22.865		16.352		23.816				
Breaking Current (kA, rms, symm)		:		11.555		8.264		12.036				
Steady State Current (kA, rms)		:	2.693	11.555		8.264		12.036				

Indicates a fault current contribution from a three-winding transformer.

* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: **Bus Sea Water**
Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus Sea Water	Total	0.00	4.165	0.00	105.58	119.83	3.611	3.611	3.01E+003	1.70E+003	4.76E+003	1.51E+003
LV Switchrack	Bus Sea Water	77.56	4.036	82.17	100.30	109.53	3.555	3.611	3.14E+003	1.70E+003	4.76E+003	1.51E+003
Sea Water	Bus Sea Water	105.00	0.150	105.00	105.00	105.00	0.069	0.000	4.81E+004	8.34E+004		
Bus Heater	LV Switchrack	77.56	0.000	82.17	100.30	109.53	0.000	0.000				
Bus Lighting P.	LV Switchrack	77.56	0.000	82.17	100.30	109.53	0.000	0.000				
Bus Crane	LV Switchrack	80.95	0.453	84.04	100.79	110.15	0.248	0.000	4.70E+003	1.31E+004		
Bus HT	LV Switchrack	77.56	0.000	82.17	100.30	109.53	0.000	0.000				
Bus UPS	LV Switchrack	77.56	0.000	82.17	100.30	109.53	0.000	0.000				
Bus S. Trafo	LV Switchrack	80.23	3.036	85.67	98.29	109.99	3.009	3.611	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	77.56	0.000	82.17	100.30	109.53	0.000	0.000				
Bus Gel Breaking	LV SW2	79.92	0.548	83.46	100.54	110.03	0.299	0.000	3.55E+003	1.09E+004		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)	:	4.165		3.611		3.544		4.156				
Peak Current (kA), Method C	:	6.047		5.244		5.146		6.034				
Breaking Current (kA, rms, symm)	:			3.611		3.544		4.156				
Steady State Current (kA, rms)	:	2.382		3.611		3.544		4.156				

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus:

Bus UPS

Nominal kV

= 0.400

Voltage c Factor

= 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus UPS	Total	0.00	5.502	0.00	100.97	119.01	5.031	5.031	2.08E+003	1.58E+003	3.09E+003	1.28E+003
LV Switchrack	Bus UPS	63.43	5.502	69.56	98.23	110.29	5.031	5.031	2.08E+003	1.58E+003	3.09E+003	1.28E+003
Bus Heater	LV Switchrack	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus Lighting P.	LV Switchrack	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus Crane	LV Switchrack	67.99	0.608	72.21	98.76	111.31	0.349	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	65.05	0.086	70.37	98.45	110.76	0.042	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus S. Trafo	LV Switchrack	67.06	4.075	74.54	95.57	110.67	4.220	5.031	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	63.43	0.000	69.56	98.23	110.29	0.000	0.000				
Bus Gel Breaking	LV SW2	66.60	0.735	71.39	98.45	111.08	0.421	0.000	3.55E+003	1.09E+004		

		3-Phase	L-G	L-L	L-L-G
Initial Symmetrical Current (kA, rms)	:	5.502	5.031	4.690	5.681
Peak Current (kA), Method C	:	8.094	7.401	6.899	8.357
Breaking Current (kA, rms, symm)	:		5.031	4.690	5.681
Steady State Current (kA, rms)	:	2.511	5.031	4.690	5.681

Indicates a fault current contribution from a three-winding transformer.

* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer.

Fault at bus: Gen Bus Power Supply
Nominal kV = 6.000
Voltage c Factor = 1.10 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Gen Bus Power Supply	Total	0.00	1.018	0.00	104.10	106.55	1.122	1.122	1.72E+002	1.03E+003	1.43E+002	6.74E+002
Bus Mkomp	Gen Bus Power Supply	0.75	0.267	0.49	103.97	106.81	0.176	0.000	1.02E+003	3.84E+003		
Bus P. Trafo	Gen Bus Power Supply	0.12	0.156	0.08	104.07	106.59	0.107	0.000	2.51E+003	6.32E+003		
Bus GTG	Gen Bus Power Supply	2.21	0.603	4.09	104.47	105.10	0.841	1.122	1.24E+002	1.75E+003	1.43E+002	6.74E+002
Motor Kompessor	Bus Mkomp	110.00	0.267	110.00	110.00	110.00	0.176	0.000	9.90E+002	3.83E+003		
Bus S. Trafo	Bus P. Trafo	12.43	0.156	68.02	63.44	115.93	0.107	0.000	2.50E+003	6.32E+003		
GTG	Bus GTG	110.00	0.603	110.00	110.00	110.00	0.841	1.122	9.14E+001	1.74E+003	9.14E+001	6.40E+002
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:	1.018	1.122		0.856		1.107				
Peak Current (kA), Method C		:	2.380	2.623		2.001		2.589				
Breaking Current (kA, rms, symm)		:		1.122		0.856		1.107				
Steady State Current (kA, rms)		:	0.195	1.122		0.856		1.107				

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: LV SW2
Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
LV SW2	Total	0.00	9.390	0.00	95.05	108.17	10.140	10.140	5.09E+002	1.45E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus Gel Breaking	LV SW2	5.44	1.254	3.68	94.27	110.09	0.849	0.000	3.55E+003	1.09E+004		
Bus Heater	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus Lighting P.	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus Crane	LV Switchrack	8.08	1.038	5.47	94.32	110.96	0.703	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	2.83	0.147	1.64	94.99	109.19	0.085	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus UPS	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus S. Trafo	LV Switchrack	8.35	6.955	14.91	92.66	105.96	8.505	10.140	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Gel Breaking (F)	Bus Gel Breaking	105.00	1.254	105.00	105.00	105.00	0.849	0.000	2.97E+003	1.08E+004		
Pedestal Crane	Bus Crane	105.00	1.038	105.00	105.00	105.00	0.703	0.000	3.70E+003	1.27E+004		
Sea Water	Bus Sea Water	105.00	0.147	105.00	105.00	105.00	0.085	0.000	4.81E+004	8.34E+004		
Bus P. Trafo	Bus S. Trafo	44.84	6.955	65.62	104.27	80.23	8.505	10.140 *	5.76E+002	1.83E+003	4.13E+002	6.19E+002
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:	9.390	10.140		7.842		10.552				
Peak Current (kA), Method C		:	18.208	19.662		15.206		20.460				
Breaking Current (kA, rms, symm)		:		10.140		7.842		10.552				
Steady State Current (kA, rms)		:	2.665	10.140		7.842		10.552				

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Fault at bus: LV Switchrack
Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
LV Switchrack	Total	0.00	9.390	0.00	95.05	108.17	10.140	10.140	5.09E+002	1.45E+003	5.86E+002	9.33E+002
Bus Heater	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus Lighting P.	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus Crane	LV Switchrack	8.08	1.038	5.47	94.32	110.96	0.703	0.000	4.70E+003	1.31E+004		
Bus Sea Water	LV Switchrack	2.83	0.147	1.64	94.99	109.19	0.085	0.000	5.07E+004	8.37E+004		
Bus HT	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus UPS	LV Switchrack	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus S. Trafo	LV Switchrack	8.35	6.955	14.91	92.66	105.96	8.505	10.140	6.85E+002	1.96E+003	5.86E+002	9.33E+002
Bus AUX	LV SW2	0.00	0.000	0.00	95.05	108.17	0.000	0.000				
Bus Gel Breaking	LV SW2	5.44	1.254	3.68	94.27	110.09	0.849	0.000	3.55E+003	1.09E+004		
Pedestal Crane	Bus Crane	105.00	1.038	105.00	105.00	105.00	0.703	0.000	3.70E+003	1.27E+004		
Sea Water	Bus Sea Water	105.00	0.147	105.00	105.00	105.00	0.085	0.000	4.81E+004	8.34E+004		
Bus P. Trafo	Bus S. Trafo	44.84	6.955	65.62	104.27	80.23	8.505	10.140 *	5.76E+002	1.83E+003	4.13E+002	6.19E+002
Gel Breaking (F)	Bus Gel Breaking	105.00	1.254	105.00	105.00	105.00	0.849	0.000	2.97E+003	1.08E+004		
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)	:	9.390		10.140		7.842		10.552				
Peak Current (kA), Method C	:	18.208		19.662		15.206		20.460				
Breaking Current (kA, rms, symm)	:			10.140		7.842		10.552				
Steady State Current (kA, rms)	:	2.665		10.140		7.842		10.552				

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Short-Circuit Summary Report

3-Phase, LG, LL, LLG Fault Currents

Bus		3-Phase Fault			Line-to-Ground Fault				Line-to-Line Fault				*Line-to-Line-to-Ground			
ID	kV	I"k	ip	Ik	I"k	ip	Ib	Ik	I"k	ip	Ib	Ik	I"k	ip	Ib	Ik
Bus AUX	0.400	0.164	0.237	0.164	0.137	0.198	0.137	0.137	0.142	0.205	0.142	0.142	0.154	0.222	0.154	0.154
Bus Crane	0.400	6.682	11.118	2.534	5.942	9.887	5.942	5.942	5.607	9.330	5.607	5.607	6.772	11.267	6.772	6.772
Bus Gel Breaking	0.400	7.943	13.683	2.606	7.663	13.200	7.663	7.663	6.650	11.455	6.650	6.650	8.535	14.702	8.535	8.535
Bus GTG	6.000	1.018	2.413	0.195	1.140	2.703	1.140	1.140	0.857	2.032	0.857	0.857	1.114	2.640	1.114	1.114
Bus Heater	0.400	3.982	5.772	2.371	3.496	5.067	3.496	3.496	3.415	4.949	3.415	3.415	3.962	5.742	3.962	3.962
Bus HT	0.400	5.502	8.094	2.511	5.031	7.401	5.031	5.031	4.690	6.899	4.690	4.690	5.681	8.357	5.681	5.681
Bus Lighting P.	0.400	3.171	4.578	2.253	2.740	3.956	2.740	2.740	2.728	3.938	2.728	2.728	3.117	4.500	3.117	3.117
Bus Mkomp	6.000	1.015	2.337	0.195	1.111	2.558	1.111	1.111	0.853	1.963	0.853	0.853	1.109	2.555	1.109	1.109
Bus P. Trafo	6.000	1.017	2.363	0.195	1.119	2.600	1.119	1.119	0.855	1.986	0.855	0.855	1.108	2.574	1.108	1.108
Bus S. Trafo	0.400	9.877	19.543	2.693	11.555	22.865	11.555	11.555	8.264	16.352	8.264	8.264	12.036	23.816	12.036	12.036
Bus Sea Water	0.400	4.165	6.047	2.382	3.611	5.244	3.611	3.611	3.544	5.146	3.544	3.544	4.156	6.034	4.156	4.156
Bus UPS	0.400	5.502	8.094	2.511	5.031	7.401	5.031	5.031	4.690	6.899	4.690	4.690	5.681	8.357	5.681	5.681
Gen Bus Power Supply	6.000	1.018	2.380	0.195	1.122	2.623	1.122	1.122	0.856	2.001	0.856	0.856	1.107	2.589	1.107	1.107
LV SW2	0.400	9.390	18.208	2.665	10.140	19.662	10.140	10.140	7.842	15.206	7.842	7.842	10.552	20.460	10.552	10.552
LV Switchrack	0.400	9.390	18.208	2.665	10.140	19.662	10.140	10.140	7.842	15.206	7.842	7.842	10.552	20.460	10.552	10.552

All fault currents are in rms kA. Current ip is calculated using Method C.

* LLG fault current is the larger of the two faulted line currents.

Sequence Impedance Summary Report

Bus		Positive Seq. Imp. (ohm)			Negative Seq. Imp. (ohm)			Zero Seq. Imp. (ohm)			Fault Zf (ohm)		
ID	kV	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance
Bus AUX	0.400	1.47738	0.03436	1.47778	1.47768	0.03628	1.47813	2.34527	0.04110	2.34563	0.00000	0.00000	0.00000
Bus Crane	0.400	0.02060	0.02987	0.03629	0.02149	0.03208	0.03862	0.03661	0.03182	0.04850	0.00000	0.00000	0.00000
Bus Gel Breaking	0.400	0.01542	0.02635	0.03053	0.01608	0.02840	0.03263	0.02541	0.02124	0.03312	0.00000	0.00000	0.00000
Bus GTG	6.000	0.59152	3.69647	3.74350	0.61692	3.90499	3.95342	0.32914	2.30399	2.32738	0.00000	0.00000	0.00000
Bus Heater	0.400	0.05323	0.02956	0.06089	0.05353	0.03148	0.06210	0.08087	0.02891	0.08589	0.00000	0.00000	0.00000
Bus HT	0.400	0.03509	0.02666	0.04407	0.03539	0.02859	0.04549	0.05203	0.02156	0.05632	0.00000	0.00000	0.00000
Bus Lighting P.	0.400	0.07046	0.02974	0.07647	0.07076	0.03166	0.07752	0.10826	0.02936	0.11217	0.00000	0.00000	0.00000
Bus Mkomp	6.000	0.66522	3.69498	3.75438	0.71310	3.92291	3.98720	0.66838	2.46794	2.55685	0.00000	0.00000	0.00000
Bus P. Trafo	6.000	0.63773	3.69194	3.74661	0.67694	3.91543	3.97352	0.55860	2.43419	2.49746	0.00000	0.00000	0.00000
Bus S. Trafo	0.400	0.00763	0.02334	0.02455	0.00787	0.02507	0.02627	0.00694	0.01042	0.01252	0.00000	0.00000	0.00000
Bus Sea Water	0.400	0.05067	0.02869	0.05823	0.05190	0.03067	0.06028	0.08015	0.02546	0.08409	0.00000	0.00000	0.00000
Bus UPS	0.400	0.03509	0.02666	0.04407	0.03539	0.02859	0.04549	0.05203	0.02156	0.05632	0.00000	0.00000	0.00000
Gen Bus Power Supply	6.000	0.61893	3.69116	3.74269	0.65736	3.91384	3.96866	0.51643	2.42527	2.47965	0.00000	0.00000	0.00000
LV SW2	0.400	0.00857	0.02436	0.02582	0.00887	0.02628	0.02774	0.00986	0.01570	0.01854	0.00000	0.00000	0.00000
LV Switchrack	0.400	0.00857	0.02436	0.02582	0.00887	0.02628	0.02774	0.00986	0.01570	0.01854	0.00000	0.00000	0.00000